

Shyantant Kundu

UG STUDENT, B. TECH CSE (3RD YEAR, PURSUING)

Objective

Embedded Systems & IoT Engineer

With all my due skillset I want to solve problem statements lying around in the world for the future betterment of society and personal development as well. I desire to engineer solutions in fields of green & alternative energy, recycling & Electronics, IoT based products for usage in daily lives.

Education

2022-Current	Institute of Engineering & Management (IEM) Kolkata B. Tech in Computer Science Engineering - Pursuing, 3rd Year - Average CGPA: 9.62 2nd Year YGPA: 9.54, 1st Year YGPA: 9.83, 5th Semester CGPA: 9.36
2020-2022	Calcutta Airport English High School (H.S.) 10+2, WBCHSE - Physics, Chemistry, Mathematics + Computer Sc. - Score: 91.40 %, Rank: 2nd in Science Stream of School
2008-2020	Calcutta Airport English High School (H.S.) 10th Boards, WBBSE - Subjects – PCM, Arts, English, Hindi - Score: 91.42 %, Rank: In School Top 10

Accomplishments

May 2025	WINNER – UST SIGHT 2.0 Grand Finale – UST Global Campus, Trivandrum Declared as 1st Position Winner at UST SIGHT 2.0, the national innovation challenge organized by UST COLOR GOLD, among 14 finalists shortlisted from 1000+ project entries across 300+ professional colleges in India. Team Oillovators from IEM Kolkata was awarded ₹50,000, certificates, and mementos for presenting the Waste Oil Recycler under the theme of sustainable innovations. The team excelled in the Boardroom Presentation and Stall Execution rounds.
April 2025	WINNER – Biznez Plan, Srijan'25 – Jadavpur University Secured 1st Position in the Business Plan Competition at Jadavpur University's Annual Techno-Management Fest Srijan'25, among 16 finalists selected from 98 applicants, for presenting the Waste Oil Recycler as a sustainable entrepreneurial innovation. 1st Runner-Up – Innovate For Impact 2025 – Loreto College & IIM Calcutta Innovation Park Secured 2nd position among the Top 8 finalists selected from 40+ teams at Loreto House College for presenting the Waste Oil Recycler business proposal; awarded ₹20,000 in the presence of Chief Guest Mr. Derek O'Brien, judged by a 4-member expert panel.
February 2025	2nd - Best Poster Presentation Award - International Conference on Electronics, Materials Engineering and Nano Technology, 2025
November 2024	WINNER – AICTE Inventors' Challenge 2024 – LIET, Noida Declared as 1st Position Winner in Inventors' Challenge 2024 conducted by All India Council for Technical Education and STMicroelectronics among Top 100 out of 2765 applicants Nationally for Automating "Waste Oil Recycler" with STM-32 Industrial Micro-controller. WINNER – Model presentation Competition, Industry Academia Conclave 2024 - NIT Jamshedpur WINNER – Innovate-ON - IET IEM Students Chapter, EE & EEE Dept. of IEM 3rd – Pitchers 2024 – Indian Institute of management, Calcutta
October 2024	2nd – Start-Up Conclave – Nirmiti 2024 RERF, Barrackpore
September 2024	2nd – Pitch Perfect – Institute of Chemical Technology, Bhubaneswar
August 2024	3rd – Bio-Informatics Big Data Analytics Hackathon – IEDC CSE, IEM
July 2024	WINNER – Start-Up Expo & Pitch Perfect – Bengal E-Summit 2024, IEM Best Poster Presentation Award - International Conference on Artificial Intelligence & Sustainable Computing (AISC 2024)
June 2024	AIR 4 – SUSTAINATHON – Indian Oil Corporation Limited Stood as Overall Runner Up among Top 12 National Finalist in IOCL's Sustainathon 2024 out of 32,000 teams from all over India for presenting Waste Oil Recycler as a solution under Extended producer responsibility (EPR) theme.
March 2024	WINNER – Green Bengal Innovation Contest 2023-24 Declared as 1st in Green Bengal Innovation's contest Pitch-to-Jury under Young Innovators Programme among 250 selected applicants with my team and Project Waste Oil Recycler. WINNER - Innovare, INNOVACION 2024 (Annual College Tech-Fest) 1st Runner Up, Electrosides, INNOVACION 2024

	February 2024 August 2023	5th Rank in CSE, Rank Certificate Ceremony, IEM • WINNER – Green technology track, IEEE WIE Hackathon 2023 1st position in the Green Technology based Sustainability track with the solution of Tribo-electric Nano-generator in IEEE WIE Hackathon conducted by IEEE Kolkata Section organized at Jadavpur University as a Team.
	July 2023	• Global Jury Validation and RV Certificate, Wadhwani NEN Received Certification of Practice Entrepreneurial Venture as Real Venture.
	April 2023	• 2nd - Ideathon, Elevate 2023 (IEEE SB Event) Innovative Project Exhibition in Annual College Tech-Fest.
	March 2023	• 1st Runner Up, INNOVACION 2023 (Annual College Tech-Fest) Won 2 nd prize in Innovative Project Exhibition for presenting my project Plastic Refill Recycler
		• Special Project & Poster Recognition, INNOVACION 2023 • 4 Technical projects exhibited at Project Exhibition & Poster Presentation.
Work experience	March, 2024 – Current	IIFR Lab, (IEM Institute of Fundamental Research), IEM, Sec V, Kolkata Student Researcher, Mentor – Dr. Priti Das (Ph.D., Lab Researcher) Projects – Automation of 2 nd version of “Waste Oil (Lubricant) Recycler” Primary work – - Scientific Research on Recycling & Realization of scientific experimental step and setup into a modular and scalable prototype - Micro-controller coding, Developing algorithm for Machine automation - Sensor, Actuator & Controller integration & Electrical circuit design of the prototype - Part designing & fabrication assist Skills acquired – - Practical Research to Working Prototype development under Hard Deadline - Industrial Automation using STM32 Micro-controller - Preparing Pilot plans for Business deployment and future scalability of the Project - Parts, Material Finding & Acquisition
	March, 2023 – Dec, 2023	IEDC Lab, (Institutional IIC), IEM, Sec V, Kolkata Student Member, Mentor – Dr. Priti Das (Ph.D., Lab Researcher) Projects – - Magnetic Field Sensing Anti-Theft Luggage Security – Tag and Tracker - Waste Oil (Lubricant) Recycler Primary work – - Electrical circuit & Mechanical part design of newly conceptualized machine, - Led Teams successfully with mentorship developing different Prototypes - Research, report and Patent-documenting Skills acquired – - Discovering Real Life Problem Statements - Designing real-life solutions around Problem Statements - Building projects from scratch to full prototypical level
	May, 2023 – July, 2023	AWS Academy, Amazon AI-ML Virtual Internship Skills Acquired – - Completed Cloud & Machine Learning Foundation Course to understood basic concepts - Learnt to operate AWS Dashboard and Sagemaker
Publications	April, 2025	Co-Author, “ Spatial Pattern Analysis of Aerial Imagery with Improved Pyramid Scene Parsing Network ” Presented at the 3rd International Conference on Computational Intelligence, Data Science and Cloud Computing (IEM-ICDC 2025), organized by Institute of Engineering & Management, Kolkata, held on April 11–12, 2025 Authors – Shyantant Kundu, Adrija Ghosh, Apratim Haldar, and Dr. Anindita Das Bhattacharjee
	February, 2025	Author, “ Designing and Implementing an IoT-Based Long-Distance Fire-Extinguishment System "Stationary 3-Axis Fire-Emergency Response Base" " ” - Authors- Shyantant Kundu, Swarajit Dey, Md. Mustafa Abdullha, Dr. Gobinda Sen and Dr. Malay Gangopadhyay - Link to Publication: https://doi.org/10.1109/IEMTech65115.2025.10959508 International Conference on Electronics, Materials Engineering and Nano Technology, Jan 31-Feb 2, 2025
	December, 2024	Author, “ Improving Driving Information Interfaceability among vehicles through an Online V2V Dashboard ” - Authors- Shyantant Kundu, Koyelisha Khan and Dr. Sayan Sarkar - PAPERID-1571087416 (Presented in Communications Track of IRTM Conference) Interdisciplinary Research in Technology & Management, Dec 5-7, 2024

July, 2024

Co-Author, **"Testing Applicability of PSPNet for Semantic Edge Detection and Segmentation of Satellite Images"**

- Authors- Adrija Ghosh, Shyantant Kundu, Apratim Haldar and Anindita Das Bhattacharjee
 - PAPERID-83 (Presented in Machine Learning & Sustainable Computing Track)
- Artificial intelligence & Sustainable Computing, July 11-13, 2024**

Patents

2023 (Processing) **"WASTE MACHINE OIL AND LUBRICANT RECYCLER "**,

Name Of Inventors - Shyantant Kundu, Adrija Ghosh, Suvojit Maity, Dr. Priti Das, Arghya Roy, Souhridya Bhattacharjee, Dibakar Roy Chowdhury, Dr. Satyajit Chakrabarti

2023 (Filed) **"MAGNETIC FIELD SENSING ANTI-THEFT LUGGAGE SECURITY – TAG AND TRACKER "**

Indian Patent Office, Appl. No.- 202431013033; Name Of Inventors - Shyantant Kundu, Adrija Ghosh, Suvojit Maity, Dr. Priti Das, Dibakar Roy Chowdhury, Dr. Satyajit Chakrabarti

2022-2023
(Published) **"PLASTIC REFILL RECYCLER "**

Indian Patent Office, Appl. No.- 202331034181; Name Of Inventors - Shyantant Kudu, Suvojit Maity, Soumalya Maiti, Sourish sarkar, Soham De, Subhradip Bal, Dr. Chandan Adhikari, Dr. Prabir Kumar Das

Computer skills

Programming

- Problem Solving, Data Structures, OOP,
 - Micro-controller coding: Arduino, ESP32, STM32
 - C, (Mastered), C++, Python, Arduino.CC (Advanced)
- Languages
- Software Skills
- MS Office – Word, PPT, Excel, Access
 - DBMS – MySQL,
 - Webpage development – HTML,
 - Simulation – MATLAB, Xilinx, Logisim, LTSpice, TinkerCad
 - Digital Designing – Canva

Hard Skills

Hardware

- Developing physical projects: Electrical circuits, Mechanical part
 - Operating Power tools & Technical Measuring Instruments
 - Machine & Product Hand-drawn Blueprint
 - Intricate design of Newly conceptualized machine and its internal parts
- Designing
- IT Hardware Skills
- Managing & Repairing PC Systems

Software/IoT Projects

Role

Team Lead & Front-end Dev

- **Online Dashboard for Vehicle-to-Everything Networking**, 3rd Yr Project
 - A 5 page Multi-suit of features for Inter-Vehicle Communication and Easy Driving Information Interfaceability
 - Scientific Research Paper published on same subject

Team Lead & Project Ideation (Micro-controller based)

- **Automation for Waste Oil Recycler**, 3rd Yr Project
 - STM-32 Micro-controller Coding for Complete Machine Automation.
- **Magnetic Lock Sensor**, 2nd Yr Project
 - An IoT Prototype built in regards of Anti-Theft Luggage Security-Tag powered by Bluetooth HC-05 running on Arduino-interface
 - Patent Applied under Indian Patent office

IoT Device Dev Lead

- **Heart Rate Monitor**, 2nd Yr Winter Project
 - A low-cost IoT Heart Health Monitoring Device enabled with an ML model for Heart Disease prediction with multi-parameter input
 - Achieved 3rd position in Bio-informatics hackathon of College

Scrum Master

- **E-Commerce website for Pizza Delivery**, 3rd Yr Project
 - Coordinated Daily Scrum, Practiced Agile Developmental Strategies
 - Managed whole team while working as a Designer for Front-end

Solo projects

- **Number Guessing App**, 2nd Yr Project
 - A fun and tactical Game with multi-parameter inputs for guessing a Number with player settings as User v/s System or User v/s user
- **Multi-function Calculator**, 2nd Yr Project
 - A Robust 12-function Mathematical Calculator for basic-to-intermediate calculation needs

Hardware Projects

Role	Team Lead & Project Ideation	• Waste Oil Recycler, Summer Project of 2nd Yr & 1st Yr ○ This project deals with Hazardous Waste Recycling Sector for recycling Used Lubricant by conversion into a low-cost Fuel-oil. ○ 2 prototypes developed with Latest prototype being Fully Functional and Automated Operation. ○ Patent Applied under Indian Patent office. ○ Won Multiple Achievements under National-level Innovative Prototype and Green Business Idea Competitions • Stationary Three-Axis Fire Emergency Response Base, 2nd Yr Project This project serves Fire-safety sector by recognizing Coordinates of the site-of-fire and Aligning itself to shoot calculated Projectile of Retardant as per the Fire-hazard • Plastic Refill Recycler, 1st Yr Project ○ A project made in the field of In the field of Recycling Sector recycling empty plastic refills into a low-cost re-usable 3-D filament ○ Patent Applied under Indian Patent office ○ Won Multiple College Innovative-Project Competitions • Stress Sense Pedal, 1st year Winter Project Device for reducing Stress-induced Anxiety by sensing Stress from repetitive Foot-tapping
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Professional Certifications

May, 2025	• NPTEL Online Certification – Industrial IoT: 86% (Elite Gold)
May, 2025	• NPTEL Online Certification – Introduction to Machine Learning: 64%
May, 2024	• NPTEL Online Certification – Computer Networking: 75% (Elite)
May, 2024	• NPTEL Online Certification – Computer Vision & Image Processing: 49%
April, 2023	• NPTEL Online Certification – Database Management System: 61% (Elite)
April, 2023	• NPTEL Online Certification – Electric Vehicles (Part 01): 65% (Elite)

Languages

- English Working Language, Affluent Speaker, Professional Reading and Writing
- Hindi Active Listener, Fluent Speaker, Advanced Reading and Writing
- Bengali Native Language, Fluent Speaking, Intermediate reading

Personal Details & Family Background

Parents:

Father – **Susanta Kundu**

- Service – Hotel industry
- Education – Graduation in Bengali Honors,
- Hotel Management from IIAS, Kolkata

Mother – **Tanurima Kundu**

- Service – Housewife
- Education – Graduation

D.O.B: **31.03.2004** Age: **19** Gender: **Male**

City of Residence: **Kolkata**, West Bengal, India

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